Satellite Video Event of Interest Detection Using Deep Spatiotemporal Metric

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***Abstract*— The rapid advancements in video satellite technology have enabled dynamic Earth observation. However, the unique characteristics of satellite video data, such as extensive spatial coverage, sparse motion information, and significant foreground–background imbalance, introduce numerous challenges for practical applications. To address these difficulties, the task of event of interest (EOI) detection has emerged as a critical solution by extracting meaningful regions containing dynamic information from satellite videos. In this article, a novel deep learning framework for EOI detection, integrating spatiotemporal slice analysis and metric learning to tackle these challenges is proposed. The framework employs spatiotemporal slice analysis to convert 3-D motion information into 2-D motion trajectories, preserving essential dynamic information while minimizing background redundancy. A ResNet-based network is adopted for robust feature extraction, and a custom Earth mover's distance (EMD) metric is introduced to enhance detection precision, enabling accurate differentiation between EOIs and non-EOIs in satellite video data. Experimental results demonstrate the superior performance of the proposed method.**

***Index Terms*— Deep metric learning, Earth mover's distance (EMD), event of interest (EOI) detection, satellite video, spatiotemporal slice analysis.**

I. Introduction

RECENT advancements in aerospace technology have enabled satellites to capture continuous, high-resolution video of Earth over large areas with sustained gaze, providing unprecedented observational capabilities. This continuous imaging capability has become a valuable tool across various domains, supporting applications in transportation monitoring, security, disaster response, resource management, and environmental protection [1], [2]. The advent of satellite video technology has attracted significant research interest. Current studies focus on key areas such as object detection [3], [4], [5], [6], tracking [7], [8], [9], [10], [11], [12], [13], scene classification [14], [15], and enhancing spatial resolution [16], [17], [18] to improve the subsequent analysis.

However, the unique characteristics of satellite video data present significant challenges for existing methods in practical applications. Modern satellite platforms, such as Jilin-1, capture extensive scenes spanning approximately 19 × 4.5 km at a spatial resolution of 0.92 m over 60-s intervals. The substantial spatial coverage results in frames approaching 20 000 × 5000 pixels. Within these expansive scenes, typical objects of interest, such as vehicles and ships, are represented by only a few pixels, creating an extreme foreground–background imbalance where the regions of interest containing target motion information comprise only a small portion of the overall scene. Consequently, contemporary methodologies predominantly utilize preextracted regions from these extensive scenes, as direct application to the full satellite video scene would significantly increase computational demands, leading to longer processing times and reduced accuracy. This limitation underscores the necessity for approaches that address the extensive spatial coverage and background redundancy of full satellite video scenes, enabling efficient extraction of the area of interest—specifically, those containing dynamic object information. Following established conventions in satellite video analysis [19], as shown in Fig. 1, we define events as local spatiotemporal blocks, where event of interest (EOI) specifically denote spatiotemporal blocks containing dynamic targets, while non-EOIs are characterized by static landscapes devoid of dynamic targets. Detecting EOIs allows us to focus on tasks such as target detection and tracking within spatiotemporal blocks of interest, effectively minimizing the influence of numerous non-EOIs. This targeted approach enables various current applications to operate more efficiently, achieving accurate results while maintaining practical feasibility in real-world scenarios.

Video anomaly detection [20], [21], [22], [23], [24] and video classification [25], [26], [27], [28], [29] in natural video analysis share conceptual similarities with EOI detection in satellite videos. Generally, video anomaly detection approaches rely on the assumption that the training data exclusively comprises normal instances. Anomalies are thus identified based on their deviation from a learned model of normality. Among such approaches, reconstruction-based methods [30], [31], [32] using autoencoders have gained prominence. These methods operate under the premise that autoencoders, trained solely on normal events, will fail to accurately reconstruct anomalous events, thereby signaling their abnormality. Alternatively, prediction-based methods [33], [34], [35], employing frameworks such as ConvLSTM networks or predictive coding, forecast future frames or motion patterns. In these approaches, discrepancies between predicted and actual frames indicate anomalous occurrences. However, satellite video data are fundamentally different due

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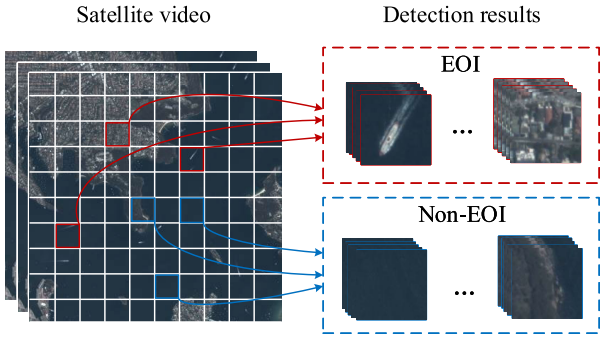


Fig. 1. EOI detection in satellite video. A local spatiotemporal block represents an event. EOIs include dynamic targets such as sailing ships and moving cars, while non-EOIs are characterized by static backgrounds lacking dynamic targets.

to their high-altitude perspectives and the extremely small scale of observable targets. These distinctive characteristics substantially limit the direct applicability and effectiveness of existing anomaly detection methods. When applied directly to EOI detection tasks, such methods often produce a high rate of false positives. In contrast, formulating satellite video EOI detection as a binary classification task—distinguishing EOIs from non-EOIs—and leveraging advanced video classification frameworks to effectively model and capture spatiotemporal features can lead to more robust and accurate detection performance. Within the satellite video analysis domain, research specifically addressing EOI detection remains limited. While Gu et al. [19] made notable progress by proposing local trajectory analysis with sparse dictionary learning under weak supervision, this approach faces certain limitations. The complex nature of target movements in satellite videos challenges the effectiveness of dictionary learning, potentially leading to detection failures. Furthermore, the method's dependence on manual optimization of dictionary parameters and support vector machine parameters introduces scalability issues and potential inconsistencies in detection performance.

In this article, we present a novel deep metric learning framework designed to advance EOI detection in satellite videos. The proposed approach integrates three key components to address the challenges inherent in this domain. First, we employ a spatiotemporal slice analysis method [36], [37], [38] that effectively projects 3-D motion information of moving targets into 2-D motion trajectories on spatiotemporal slices. This transformation captures essential motion features while eliminating redundant static background elements. Second, to address the complexity of identifying events with intricate motion trajectories, we utilize a 12-layer ResNet [39] for feature extraction from the spatiotemporal slices. This deep learning model with robust feature representation enhances the accuracy of EOI detection. Finally, we introduce a novel distance measurement function grounded in deep metric learning [40], [41], [42], specifically tailored for EOI detection. This function significantly improves detection precision and strengthens the framework's generalization capabilities across diverse scenarios. The primary contributions of this study are as follows.

1) *Pioneering the application of deep metric learning for EOI detection:* We present a novel deep metric learning framework designed specifically for detecting EOIs in satellite video data. This innovative approach effectively addresses the challenges associated with EOI identification, significantly enhancing detection performance.
2) *Enhanced efficacy through spatiotemporal slice analysis:* Utilizing spatiotemporal slice analysis, we extract critical motion features from satellite video while effectively filtering out redundant static background information. This method not only improves detection accuracy but also optimizes computational efficiency, ensuring faster and more reliable satellite video processing.
3) *An innovative distance measurement for improved detection accuracy:* We introduce a novel distance measurement function based on deep metric learning, which refines the detection process. This new metric improves detection accuracy and enhances the model's generalization capability, ensuring robust performance across diverse satellite video scenes.

The remainder of this article is structured as follows. Section II elaborates on the proposed method, detailing the integration of spatiotemporal slice analysis and deep metric learning for EOI detection in satellite videos. Section III introduces the satellite video datasets used in the study, describing their characteristics and the experimental settings applied. Section IV presents a comprehensive analysis of the experimental results, highlighting the effectiveness and robustness of the proposed framework in various satellite video scenes. Finally, Section V concludes the study by summarizing the primary contributions and discussing potential directions for future research, including enhancing generalizability and real-time application capabilities.

II. Proposed Method

A comprehensive framework designed to detect EOI within satellite video is presented in this section. As illustrated in Fig. 2, our framework is divided into two primary stages. The initial stage encompasses a spatiotemporal slice analysis method. This technique effectively translates the 3-D motion information of moving targets within the satellite video into 2-D motion trajectories on the spatiotemporal slice. This conversion not only preserves essential motion information but also streamlines it for more efficient processing. Following this, the feature extraction module is employed to extract feature vectors from the slice. We then utilize a custom-developed Earth mover's distance (EMD) [43] metric to calculate the distances between feature vectors derived from both the query and support slices. This metric is pivotal in our approach, enabling the precise classification of spatiotemporal slices as either positive or negative samples. By executing this process, our method efficiently detects EOIs from the satellite video, offering a significant advancement in satellite video analysis.

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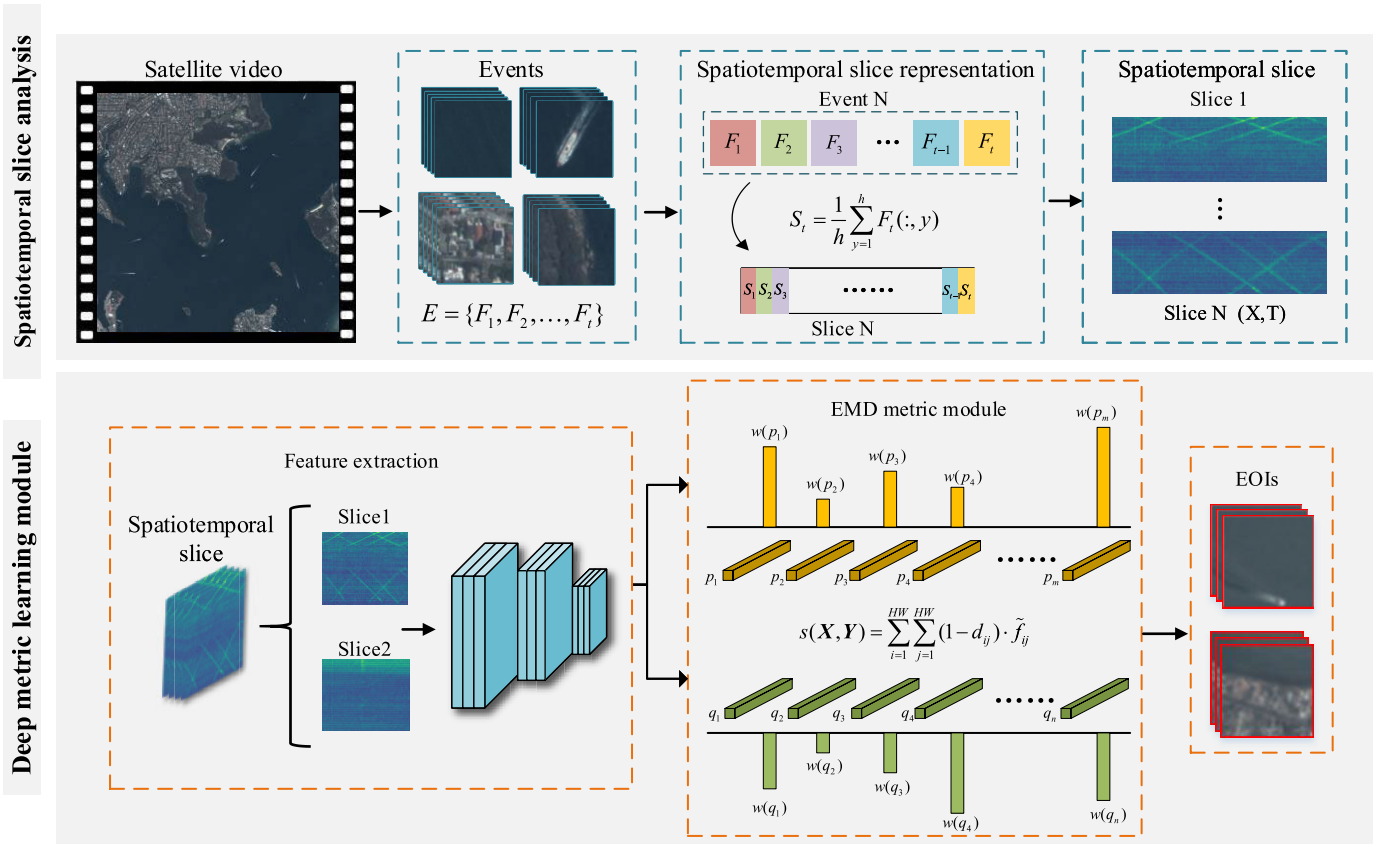


Fig. 2. Framework of the proposed method. The spatiotemporal slice analysis translates motion information from satellite videos into motion trajectories. The deep metric learning phase extracts feature vectors from the slices, utilizing a custom EMD metric to calculate distances and detect EOIs from satellite videos.

A. Spatiotemporal Slice Representation for Satellite Video

In our proposed framework, a satellite video event is represented as a sequence of frames, denoted by $E = \{F_1, F_2, \ldots, F_t\}$, where each frame F_i $(1 \leq i \leq t)$ has dimensions $h \times w$, representing the height and width of the frame. To capture the spatiotemporal characteristics of the video, each frame is first transformed into a column vector. We define a transformation operation T_f that converts each frame F_i into a column vector S_i of size $n \times 1$, where n corresponds to the number of features extracted from the frame. This transformation is expressed as

$$S_i = T_f(F_i). \tag{1}$$

The specific form of T_f depends on the feature extraction technique used, such as dimensionality reduction or encoding methods that map each frame into a feature vector.

After transforming each frame, we construct the spatiotemporal slice T, which is a matrix representing the evolution of the video event over time. This spatiotemporal slice is formed by concatenating the column vectors $S_1, S_2, \ldots, S_t$, as follows:

$$T = [S_1, S_2, \ldots, S_t] = [T_f(F_1), T_f(F_2), \ldots, T_f(F_t)]. \tag{2}$$

The resulting matrix T has dimensions $n \times t$, representing the spatiotemporal characteristics of the satellite video event. This matrix encapsulates both the spatial information within each frame and the temporal variations across the sequence of frames.

To represent the spatiotemporal structure of the video, slices can be generated in three orientations: vertical, horizontal, and diagonal. In our study, we focus on the vertical and horizontal slices, which capture key spatial and temporal relationships. The horizontal slice emphasizes the spatial distribution along the width of the frame, while aggregating information across time. To generate a horizontal slice, we first apply background subtraction to each frame F_t, which removes redundant background information and isolates the target objects. This preprocessing step ensures that only the dynamic elements of the video are considered. The processed frame F_t is then transformed into a column vector S_t as follows:

$$S_t = T_f(F_t) = \frac{1}{h}\sum_{y=1}^{h} F_t(:, y). \tag{3}$$

Here, the summation averages the pixel values along the height h of the frame, resulting in a column vector that captures the temporal evolution of spatial features across the horizontal axis. This transformation results in a horizontal slice T_H with dimensions $w \times t$, where w is the width of the frame.

In a similar manner, the vertical slice captures the spatial distribution across the height of the frame while maintaining the temporal progression. To obtain the vertical slice, we aggregate information along the width w of each frame.

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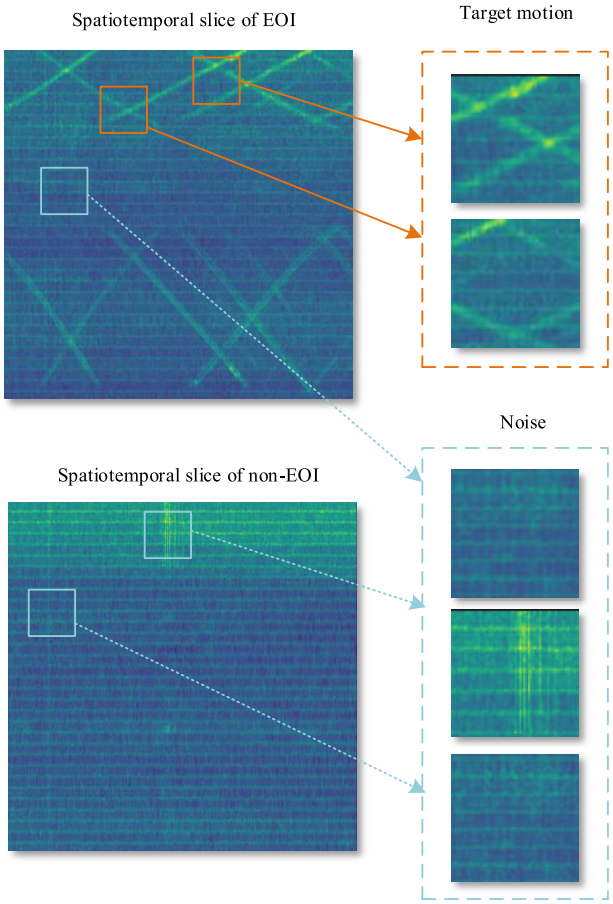


Fig. 3. Spatiotemporal slices of EOIs and non-EOIs. These slices encapsulate two primary types of information: motion information, represented as sloped lines indicating target movement, and noise artifacts, represented as horizontal and vertical stripes arising from global motion and background subtraction.

The transformation for the vertical slice is expressed as

$$S_t = T_f(F_t) = \frac{1}{w}\sum_{x=1}^{w} F_t(x,:). \tag{4}$$

This operation averages the pixel values along the width of the frame, capturing the temporal variation along the vertical axis. The resulting vertical slice T_V has dimensions $h \times t$, where h is the height of the frame.

Finally, to capture the complementary spatial information from both horizontal and vertical perspectives, we concatenate the horizontal and vertical slices. This results in the final spatiotemporal slice T, which combines the time–space X-axis and Y-axis relationships

$$T = [T_H \quad T_V]. \tag{5}$$

The final matrix T has dimensions $(w+h) \times t$, incorporating both the horizontal and vertical slices and representing the complete spatiotemporal structure of the satellite video event. By concatenating these two slices, we effectively capture the spatial and temporal dynamics of the video from complementary orientations.

As illustrated in Fig. 3, the 3-D motion information captured in satellite video is transformed into 2-D representations through the process of spatiotemporal slicing. These slices provide a compact, structured, and interpretable representation of both EOIs and non-EOIs, enabling effective EOI detection. The spatiotemporal slices encapsulate two primary types of information: target motion information and noise.

The first type of information, target motion, corresponds to the movement patterns of the objects within the satellite video. In a typical EOI scene, where specific targets are moving across the scene, their trajectories are represented as distinct lines within the spatiotemporal slices. These lines have a clear slope, which corresponds to both the direction and velocity of the target's movement. The transformation into spatiotemporal slices maps the complex 3-D motion (spatial coordinates over time) into a 2-D space, making it easier to analyze the motion. This structured formation inherently captures and preserves the temporal continuity of the target's trajectory, as persistent object movements appear consistently within adjacent columns in the slice. Importantly, these motion patterns, characterized by their linearity and distinct slope, are exclusive to EOI scenarios, making them a crucial feature for discriminating between EOIs and non-EOIs. In contrast, non-EOI scenarios, where there is either minimal movement or no movement at all, result in a lack of well-defined motion patterns within the slices.

The second type of information captured in the spatiotemporal slices is noise, which manifests as unwanted artifacts in both EOI and non-EOI scenarios. These artifacts typically manifest as horizontal or vertical stripes within the spatiotemporal slices, originating primarily from global background motion caused by the satellite's staring imaging mode or minor viewpoint adjustments, as well as potential inaccuracies introduced during frame-by-frame background subtraction. Such stripes indicate residual global motion or static background features, clearly distinct from genuine target trajectories. Although noise-related artifacts may appear in both EOI and non-EOI slices, their visual patterns notably differ from the trajectories representing actual moving objects. Consequently, our spatiotemporal slice representation not only simplifies motion representation but also facilitates the effective differentiation of meaningful dynamic information from irrelevant background noise. By employing the spatiotemporal slice representation, our method captures essential motion trajectories of moving objects, clearly encoding their movements into structured, 2-D slices. This representation significantly reduces redundant static background information, thereby streamlining data processing, enhancing detection accuracy, and improving computational efficiency.

B. Re-Examining the EMD

Through the process of spatiotemporal slice analysis, we transform the task of EOI detection from satellite video into a binary classification problem. The goal is to distinguish between EOI and non-EOI spatiotemporal slices. However, two critical challenges arise in the classification of these spatiotemporal slices that must be addressed for effective performance. The first challenge stems from the limited availability of labeled training data in satellite video analysis. Accurately labeled EOI data are often difficult and costly to obtain. This scarcity of labeled data creates the need

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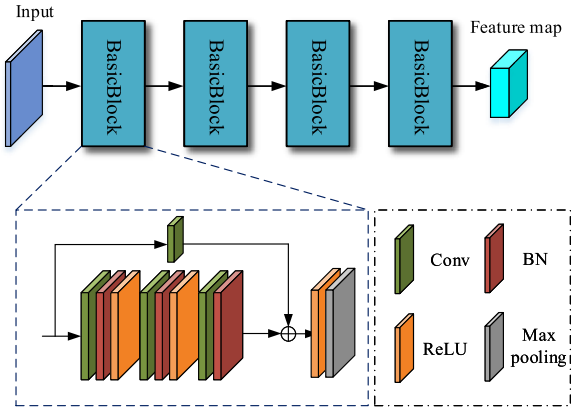


Fig. 4. Architecture of the feature extraction module.

for a model that can generalize well from limited training examples, enabling effective classification of both EOI and non-EOI spatiotemporal slices despite the small sample size. The second challenge arises from the local nature of motion information within the spatiotemporal slices. As illustrated in Fig. 3, the motion trajectories of targets typically occupy only small regions within the slices, while the majority of the slice consists of background noise. This results in a sparse distribution of discriminative features within the slice, where the meaningful motion information is localized within small portions of the slice.

To address these challenges, we propose a metric learning approach based on the EMD as the primary distance measure. This methodology is specifically designed to handle both the limited training data and the local nature of motion information in spatiotemporal slices, providing a robust framework for EOI detection in satellite video analysis. The EMD quantifies the minimum cost required to transform one distribution into another through optimal transport principles. Specifically, it determines the most efficient way to convert one probability distribution into another by computing the cost of redistributing mass between distributions. The mathematical framework of EMD can be formalized as follows.

Consider two feature sets: $P = \{p_1, p_2, \ldots, p_m\}$ and $Q = \{q_1, q_2, \ldots, q_n\}$. Each element carries a weight, denoted as $w(p_i)$ and $w(q_j)$, representing its distributional significance. The transportation cost between elements p_i and q_j is defined by a distance metric d_{ij}. The optimization problem seeks to determine the optimal flow f_{ij} between elements that minimizes the total transportation cost while satisfying distribution constraints. This can be expressed mathematically as

$$\begin{aligned} &\min \sum_{i=1}^{m}\sum_{j=1}^{n} f_{ij}d_{ij} \\ &\text{s.t.:} \sum_{j=1}^{n} f_{ij} = w(p_i) \quad \forall i \\ &\sum_{i=1}^{m} f_{ij} = w(q_j) \quad \forall j \\ &f_{ij} \geq 0 \quad \forall i, j. \end{aligned} \tag{6}$$

Here, f_{ij} represents the mass transported from p_i to q_j. The objective function minimizes the total transportation cost while ensuring weight conservation through the specified constraints.

C. EMD for Spatiotemporal Slice Metric

In our framework, each spatiotemporal slice is decomposed into a set of local representations. By assigning optimal weights to these local embeddings, we utilize EMD to compute the minimal matching cost, effectively measuring dissimilarity between slices. The implementation divides each spatiotemporal slice into a $H \times W$ grid, where each cell undergoes feature extraction through a neural network. This network encodes individual grid cells into feature vectors, collectively forming the slice's embedding set, which effectively captures localized spatiotemporal information. For comparing two spatiotemporal slices, we compute the distance between their embedding sets using EMD, following the formulation in (6). Given two embedding sets of $H \times W$ vectors, the unit cost is derived from pairwise distances between embedding nodes $\mathbf{a}_i$ and $\mathbf{b}_j$ from respective slices, where nodes with similar features yield smaller costs

$$d_{ij} = 1 - \frac{\mathbf{a}_i^T \mathbf{b}_j}{\|\mathbf{a}_i\| \|\mathbf{b}_j\|}. \tag{7}$$

The weights $w(\mathbf{a}_i)$ and $w(\mathbf{b}_j)$ are generated through a process detailed in Section II-D. The EMD computation involves solving the linear programming problem in (6), which can be efficiently implemented using either the GPU-accelerated QPTH library or OpenCV's enhanced simplex algorithm.

Given two spatiotemporal slice representations X and Y, the similarity score s is computed as

$$s(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^{HW}\sum_{j=1}^{HW}(1 - d_{ij}) \cdot \tilde{f}_{ij} \tag{8}$$

where $\tilde{f}_{ij}$ represents the optimal matching flow between nodes $\mathbf{a}_i$ and $\mathbf{b}_j$. This metric learning module generates similarity scores that effectively group spatiotemporal slices based on their relative similarities.

Our proposed method ensures optimal alignment between sparse motion features in spatiotemporal slices, effectively filtering out noise and enhancing classification accuracy. Additionally, the metric learning framework enables strong generalization performance, even when training datasets are limited, making it well-suited for EOI detection task.

D. Spatiotemporal Motion-Based Weight Generation

In the EMD formulation, node weights are crucial parameters that determine the total matching flows from each node. Nodes with larger weights exert greater influence on the comparison between two sets, while those with minimal weights contribute negligibly to the overall distance, regardless of their connectivity. For EOI detection applications, our primary objective is to identify target motion trajectories within spatiotemporal slices. Therefore, the matching algorithm should prioritize grid cells containing motion trajectories through appropriate weight assignment.

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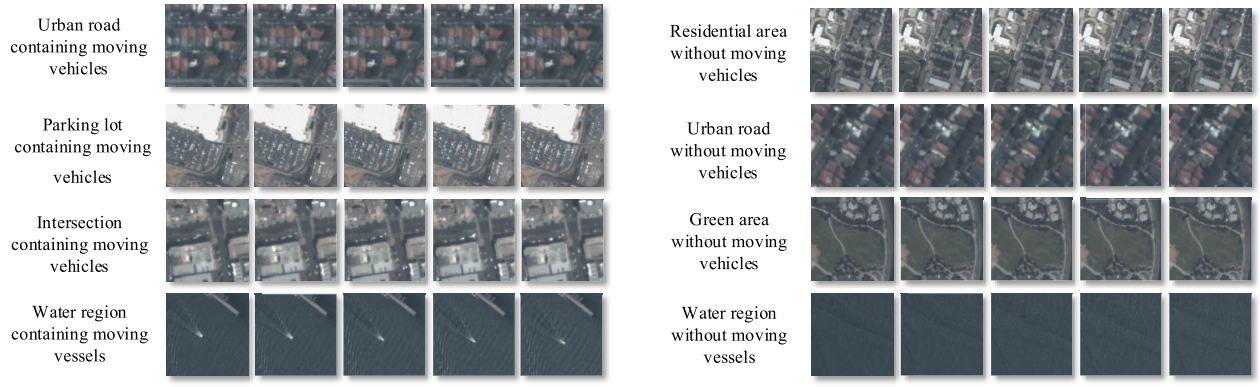


Fig. 5. Examples of EOIs and non-EOIs. (a) Frames (1, 21, 41, 61, and 81) depicting EOIs. (b) Frames (1, 21, 41, 61, and 81) depicting non-EOIs.

In spatiotemporal slices, object motion trajectories manifest as straight lines with distinct slopes, encoding both movement direction and continuity. To characterize these trajectories, we compute the motion trajectory strength for each grid cell H using

$$\mathbf{S}_{xt} = \sum_{x',t' \in w} \hat{\mathbf{H}}_x(x - x', t - t')\hat{\mathbf{H}}_t(x - x', t - t'). \quad (9)$$

Here, $\mathbf{S}_{xt}$ represents the spatiotemporal motion strength at each given grid cell within the spatiotemporal slice. $\hat{\mathbf{H}}_x = (\partial(G * H)/\partial x)$ and $\hat{\mathbf{H}}_t = (\partial(G * H)/\partial t)$ denote spatial and temporal partial derivatives of the slice H, respectively. A Gaussian kernel G is applied to the slice before tensor computation to suppress noise and ensure smooth trajectory representation. The derivatives are computed over a 3×3 window w, centered at each pixel in the spatiotemporal slice.

Subsequently, the weight W_i for each grid cell H_i is determined by

$$W_i = \sum_{p \in H_i} I(p) \quad (10)$$

where $I(p)$ denotes the intensity of pixel p within cell H_i. It is important to note that while prior work [44], [45] employed EMD metrics, our method adopts a fundamentally different approach. Instead of relying on conventional image feature weighting, we utilize spatiotemporal slice representations to encode motion trajectories into structured 2-D forms. The strength of these trajectories then guides the EMD weight distribution, assigning higher importance to regions exhibiting genuine dynamic activity, such as moving vehicles or vessels. This allows the algorithm to focus on meaningful motion while effectively suppressing static or noisy regions.

E. Implementation Details

Our model leverages a 12-layer ResNet as the feature extraction backbone, as shown in Fig. 4. ResNet12 consists of four blocks, each containing three convolutional layers with batch normalization and ReLU activation, followed by a downsampling operation using max-pooling. This architecture is well-suited for extracting fine-grained features. To tailor ResNet12 for our specific task, we remove the fully connected layer at the end of the network, enabling it to output feature maps instead of class predictions. These feature maps are aggregated into feature vectors, preserving the fine-grained local information critical for analyzing spatiotemporal slices.

TABLE I
DATASET OVERVIEW AND SPLITS

	Dataset 1		Dataset 2
	Training Set	Test Set 1	Test Set 2
EOIs	50	260	167
Non-EOIs	200	390	857
Total Events	250	650	1024

The training protocol follows a two-phase approach: initial feature pretraining followed by episodic meta-training [46]. The pretraining phase optimizes the model parameters to maximize validation accuracy, establishing a robust initialization state. The network is trained using SGD as the optimizer, with a learning rate of 0.1, momentum of 0.9, and weight decay of 5×10^{-4} for 100 epochs. Subsequently, the model undergoes refinement through episodic training, consisting of 5000 episodes. During this stage, we continue to use SGD as the optimizer, but reduce the learning rate to 4×10^{-4}, while keeping the momentum and weight decay the same as in the pretraining phase. During the evaluation phase, we construct the support set using five selected positive and negative samples from the training set that effectively represent their respective classes. The entire test set serves as the query set, where class assignments (positive or negative) for each query sample are determined through distance computations between query and support samples.

III. DATA AND EXPERIMENTAL SETTING

A. Data Description

In this study, we utilize two satellite videos from the Jilin-1 constellation. The first video comprises 300 frames, each sized 3840×3840 pixels, whereas the second video consists of 300 frames with dimensions of 4096×4096. To facilitate our analysis, both videos are divided into patches (events)

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TABLE II
AUC AND AP PERFORMANCE OF DIFFERENT METHODS ON DATASET 1 AND DATASET 2

	Dataset 1		Dataset 2	
	AUC	AP	AUC	AP
TSDEOI [19]	0.907	0.831	0.889	0.742
Swin [47]	0.916	0.872	0.802	0.598
Timesformer [48]	0.848	0.674	0.825	0.393
TANet [49]	0.876	0.791	0.752	0.491
MViT [50]	0.771	0.525	0.766	0.499
UniFormerV2 [51]	0.766	0.498	0.822	0.367
Ours	**0.942**	**0.893**	**0.937**	**0.845**

of 128 × 128 pixels, with each patch being treated as either a EOI or a nonevent of interest (non-EOI). The first video (Dataset 1) yields a total of 900 events, among which 310 are labeled as EOIs and 590 as non-EOIs. Of these, 250 events (50 EOIs and 200 non-EOIs) are designated for training, while the remaining 650 events form Test Set 1. For the second video (Dataset 2), all 1024 events (167 EOIs and 857 non-EOIs) are used exclusively for testing as Test Set 2, offering a cross-dataset evaluation to examine the generalization capability of our method. Table I provides a concise summary of the datasets, including the numbers of EOIs, non-EOIs, and how they are split into training and testing sets. As illustrated in Fig. 5, EOIs are characterized by the presence of motion information from dynamic objects such as moving vehicles and ships, and are treated as positive samples in our analysis. Conversely, non-EOIs contain primarily static backgrounds and structures without significant motion information and serve as negative samples.

B. Experimental Setting

Our evaluation framework implements a comprehensive set of performance metrics to systematically assess the effectiveness of the proposed method. The evaluation metrics encompass the receiver operating characteristic (ROC), area under the curve (AUC), precision–recall (PR) curve, and average precision (AP). The experimental protocol employs a threshold-based classification approach, where samples are categorized based on their scores relative to a predetermined threshold: samples with scores below the threshold are designated as negative, while those above are classified as positive. This classification scheme enables the computation of fundamental performance indicators: true positives (TPs), false negatives (FNs), false positives (FPs), and true negatives (TNs). The ROC curve provides a graphical representation of the classifier's discriminative capability by plotting the TP rate (TPR) against the FP rate (FPR) across varying threshold values, effectively illustrating the trade-off between sensitivity and specificity. The AUC metric quantifies the overall classification performance by computing the area beneath the ROC curve, providing a single scalar measure of effectiveness. For datasets with class imbalances, which are typical in satellite video EOI detection, the PR curve is particularly valuable. It plots precision (the ratio of TP to the sum of TP and FP) against recall (which is equivalent to TPR). This curve is especially informative when assessing models in scenarios where the positive class is underrepresented, as is often the case in satellite video event detection. The AP metric synthesizes the PR curve information by calculating the weighted mean of precisions at each threshold. This metric effectively captures the precision–recall trade-off across the threshold spectrum, proving especially relevant for imbalanced dataset scenarios. Beyond accuracy metrics, processing speed is a vital indicator of a method's practical applicability. It measures the time efficiency of the algorithm in processing satellite video data. Faster processing speeds indicate better suitability for real-time or large-scale applications, where rapid analysis is crucial.

Our comparative analysis focuses primarily on benchmarking the proposed method against the two-stream detection of events of interest (TSDEOI) approach [19], which, to our knowledge, is the only existing method specifically designed for EOI detection in satellite videos. To provide a more comprehensive evaluation, we also compare our method with techniques from the broader computer vision field. Current video anomaly detection frameworks typically rely on rich contextual information, clear object boundaries, and easily interpretable motion cues that are abundant in ground-level surveillance scenarios. However, satellite video data are fundamentally distinct, characterized by limited target visibility, sparse and subtle motion signals, and minimal spatial details due to the high-altitude perspective and small target scales. These unique attributes significantly limit the direct applicability and effectiveness of conventional anomaly detection methods. Applying them directly to EOI detection tasks often leads to a high number of false positives. Considering these specific task requirements, we have deliberately selected advanced video classification models that are capable of effectively modeling and classifying structured spatiotemporal information. For this purpose, we include five advanced video classification architectures in our comparison: the video Swin transformer (Swin) [47], Timesformer [48], temporal adaptive network (TANet) [49], MViT [50], and UniFormerV2 [51]. By comparing the performance of our method with these models, we aim to demonstrate the efficacy and superiority of our approach in the specific context of EOI detection in satellite videos. All experimental evaluations were conducted on a computing platform equipped with an NVIDIA Tesla V100-PCIE GPU.

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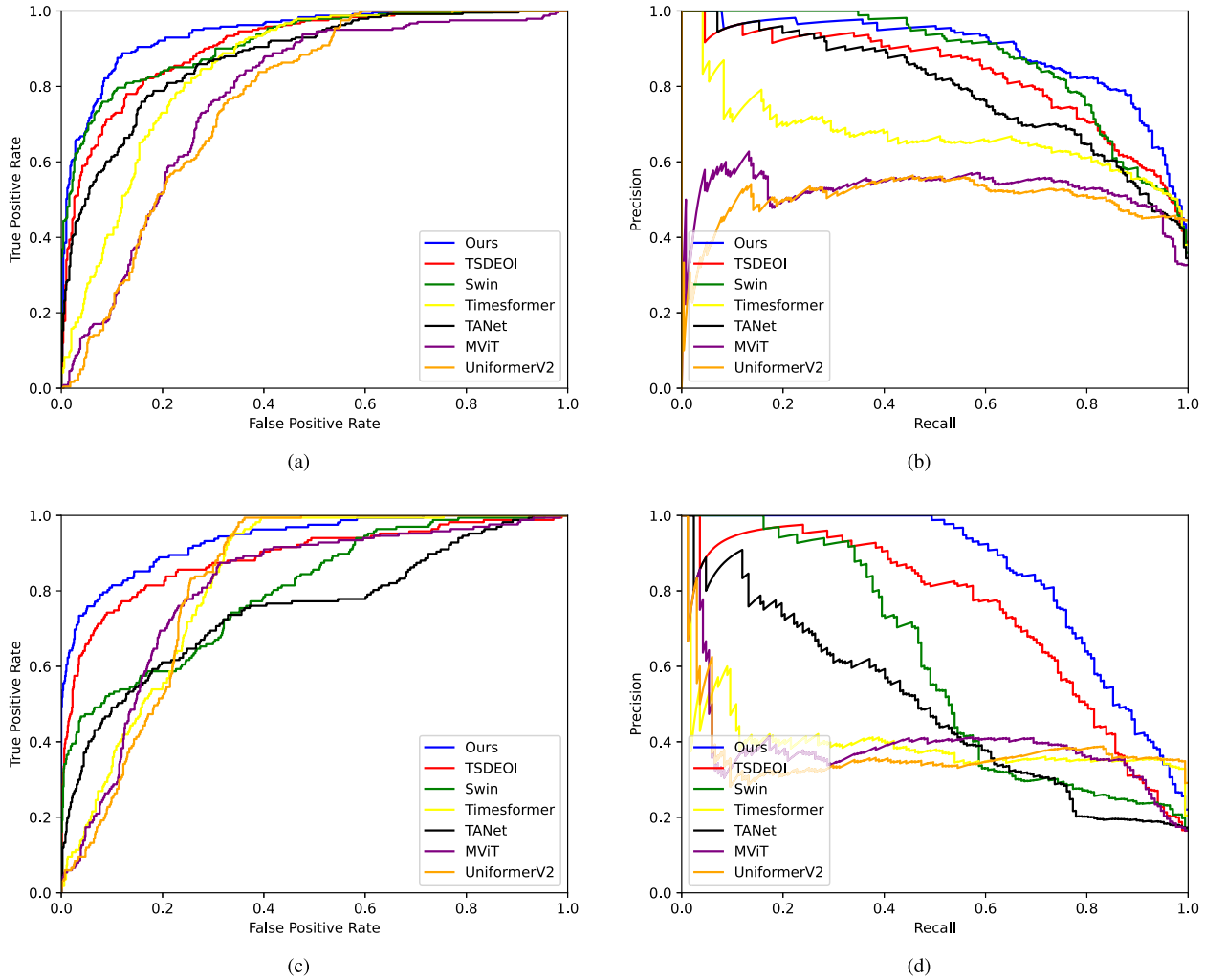


Fig. 6. ROC and PR curves of different methods for two datasets. (a) ROC curves for dataset 1. (b) PR curves for dataset 1. (c) ROC curves for dataset 2. (d) PR curves for dataset 2.

IV. Experimental Results and Analysis

A. Quantitative Performance Assessment

The performance evaluation of our proposed method was conducted through comprehensive analyses of two satellite video datasets. As shown in Fig. 6(a) and (b), these figures display the ROC and PR curves for the dataset 1. The ROC curve for the proposed method consistently surpasses the other models, particularly at lower FPRs. This indicates that the proposed approach maintains a higher TPR while minimizing the rate of false alarms, reflecting robust performance in detecting EOIs. The steep rise of the ROC curve near the origin underlines that the model effectively identifies most positive samples with few false positives. Comparative methods such as Timesformer and TANet demonstrate substantially inferior performance at equivalent FPR levels, suggesting limitations in their ability to capture the spatiotemporal characteristics of satellite video EOIs. The PR curve provides additional validation of our method's effectiveness, particularly in addressing the inherent class imbalance characteristic of satellite video EOI detection. This performance differential becomes more pronounced at higher recall rates, where conventional approaches exhibit substantial precision degradation. This robustness indicates the method's capability to maintain reliable predictions even in the face of imbalanced datasets, a common challenge in satellite video data where EOIs are relatively sparse.

The generalization capability analysis, conducted through cross-dataset validation (training on dataset 1, and testing on dataset 2), reveals the most significant advantages of our approach. As demonstrated in Fig. 6(c) and (d), our method maintains consistent performance characteristics across datasets, with minimal degradation in detection metrics. This stability contrasts sharply with baseline methods, which exhibit substantial performance deterioration. In the ROC curve, the proposed method again demonstrates superior performance, with a higher TPR across a range of FPRs compared to the baseline methods. The blue curve (Ours) retains its characteristic steep rise near the origin, which indicates strong sensitivity while maintaining specificity. In contrast, the curves for other methods, such as Swin, Timesformer, and TANet, display a flatter trajectory and lag behind in TPRs, particularly at lower FPRs. The PR curve provides a more nuanced view of the performance under imbalanced datasets. The proposed method outperforms

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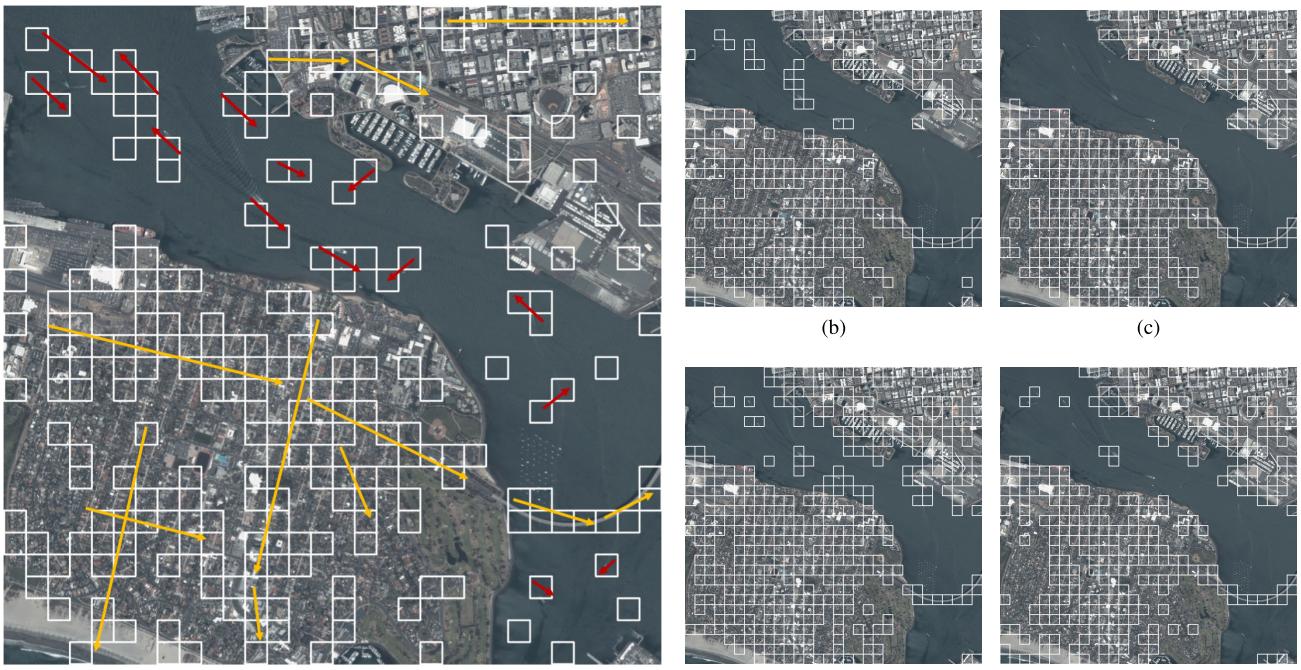


Fig. 7. Visualized results of EOI detection on Satellite Video 1. (a) Results of our method, with red arrows indicating maritime vessel movement and yellow arrows depicting land vehicle movement. (b) Results of the TSDEOI method. (c) Results of the Swin method. (d) Results of the Timesformer method. (e) Results of the TANet method.

TABLE III
AUC AND AP PERFORMANCE OF DIFFERENT WEIGHTING STRATEGIES ON DATASET 1 AND DATASET 2

	Dataset 1		Dataset 2	
	AUC	AP	AUC	AP
Weight-U	0.913	0.867	0.904	0.811
Weight-R	0.901	0.834	0.886	0.791
Ours	**0.942**	**0.893**	**0.937**	**0.845**

TABLE IV
AVERAGE EVENT PROCESSING TIME OF DIFFERENT METHODS

Method	Time
TSDEOI [19]	0.147 s
Swin [47]	0.665 s
Timesformer [48]	0.182 s
TANet [49]	0.438 s
MViT [50]	0.537 s
UniFormerV2 [51]	0.625 s
Ours	**0.119 s**

the baselines across most recall levels, maintaining higher precision even at high recall values. This ability to avoid significant drops in precision as recall increases is critical for practical applications, where retaining reliability at higher sensitivity is essential. The baseline methods show a marked decline in precision as recall increases. This decline suggests that these models produce a larger proportion of false positives when striving for higher recall, a common issue when generalization is poor. Experimental results indicate that MViT and UniFormerV2 exhibit relatively poor performance in our EOI detection task. Specifically, despite its advanced multiscale design, MViT struggles to effectively highlight subtle motion cues arising from small-scale targets in satellite videos. Similarly, UniFormerV2 encounters difficulty adapting its unified spatiotemporal representation to handle the extreme foreground–background imbalance inherent in satellite scenes.

The above experimental results demonstrate that our proposed framework effectively reduces the risk of overfitting and maintains strong generalization capabilities, even when trained on a limited number of samples. Specifically, the spatiotemporal slice representation systematically normalizes diverse and complex motion patterns into structured, consistent trajectory representations. This transformation significantly enhances both the quantity and quality of motion-related information captured within each event, enabling the network to extract robust and discriminative features independently of specific video characteristics or scene contexts. Furthermore, our custom-designed EMD metric incorporates a motion-centric weighting strategy based explicitly on the computed trajectory strength of each cell within the slices. By assigning greater importance to meaningful dynamic regions rather than irrelevant static background features, this targeted weighting approach significantly improves the model's robustness and resistance to overfitting, ultimately ensuring stable and accurate detection performance across varying satellite video datasets.

The results presented in Table II offer a detailed comparative analysis of detection performance, focusing on two key evaluation metrics: AUC and AP. Our proposed method demonstrates

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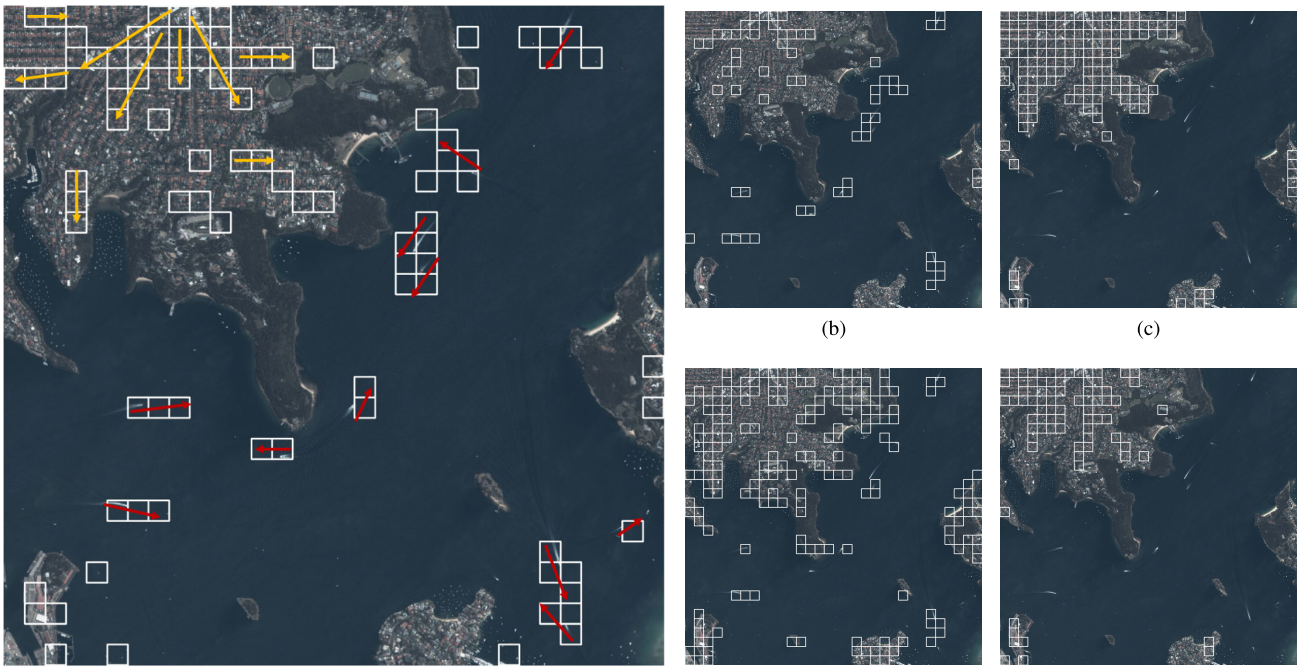


Fig. 8. Visualized results of EOI detection on Satellite Video 2. (a) Results of our method, with red arrows indicating maritime vessel movement and yellow arrows depicting land vehicle movement. (b) Results of the TSDEOI method. (c) Results of the Swin method. (d) Results of the Timesformer method. (e) Results of the TANet method.

clear superiority in both AUC and AP scores across datasets. Specifically, on the dataset 1, it achieves an AUC of 0.942 and an AP of 0.893, both of which are strong indicators of its exceptional detection capabilities. The Swin method, while showing competitive performance on dataset 1, experiences a substantial drop when tested on dataset 2. Although MViT and UniFormerV2 incorporate transformer-based structures demonstrating strong performance in general video tasks, the unique characteristics of satellite video data—including large-scale scenes, sparse motion information, and targets often represented by only a few pixels—pose substantial challenges that limit the effectiveness of these general-purpose video architectures in our specific application scenario.

The comparative analysis of EOI detection performance reveals that our proposed method not only outperforms existing models on individual datasets but also exhibits remarkable stability and robustness across different satellite video scenes. Our experimental findings indicate that the spatiotemporal slice representation and the metric learning module each play crucial and complementary roles. By converting raw satellite video frames into 2-D slices, we make subtle motion trajectories more salient, thereby reducing background noise and helping the model focus on genuinely dynamic regions. The metric learning module further refines this process by prioritizing grid cells that capture real movement and de-emphasizing uniform regions prone to global motion artifacts. As a result, the synergy between slice-based representation and EMD-driven matching significantly strengthens both accuracy and robustness when detecting EOIs in satellite video data.

To evaluate the effectiveness of the proposed motion-based weight generation component, we conduct ablation experiments comparing three weighting strategies for EMD computation: 1) our full method with motion-based weight generation; 2) uniform weighting (Weight-U), where all spatiotemporal grids are assigned equal weights; and 3) random weighting (Weight-R), where grid weights are randomly sampled and normalized. As summarized in Table III, our full method consistently outperforms both baselines on both datasets. These results demonstrate that the motion-based weight generation method provides significant performance gains over naive or random weighting schemes.

Table IV highlights the computational efficiency of various methods for detecting EOIs in satellite videos, an important consideration for real-time applications in operational satellite surveillance systems. Our proposed approach demonstrates outstanding computational efficiency, with a mean processing time of 0.119 s per event across the test samples. This computational advantage stems from our innovative transformation of the complex 3-D satellite video processing problem into a simplified 2-D spatiotemporal slice representation. By representing satellite video data as spatiotemporal slices, our method effectively reduces computational complexity without compromising detection accuracy. The spatiotemporal slice representation minimizes background redundancy, allowing the model to focus computational resources on dynamic information, which is essential for distinguishing EOIs from static background scenes in satellite videos.

The comprehensive experimental results demonstrate that our method not only achieves superior detection accuracy but also maintains robust performance across different satellite video scenarios while offering improved computational efficiency. These characteristics make it a promising solution for

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real-world satellite video EOI detection applications, where both accuracy and processing speed are crucial considerations.

B. Visualized Qualitative Analysis

Comprehensive visual comparisons of EOI detection performance across multiple methodologies are presented in Figs. 7 and 8. The visualization scheme employs white bounding boxes to demarcate detected EOIs, with motion patterns represented through a dual-color arrow system: red arrows indicating maritime vessel trajectories and yellow arrows depicting land vehicle movements. As illustrated in Figs. 7(b) and 8(b), the TSDEOI method achieves robust detection of waterborne EOIs containing moving boats. However, its performance is less reliable in terrestrial scenarios, manifesting in both increased FPRs and missed detections. This performance disparity can be attributed to two primary factors: first, the inherently higher complexity of land-based motion trajectories compared to maritime patterns; and second, the limitations of the method's feature extraction mechanism. Analysis of the Swin method, illustrated in Figs. 7(c) and 8(c), reveals critical limitations in its current implementation. The method exhibits a pronounced tendency toward over-detection in terrestrial regions, generating numerous spurious bounding boxes while simultaneously failing to identify legitimate EOIs in water regions. This weakness stems from fundamental architectural limitations: the method's heavy reliance on spatial feature learning without adequate temporal modeling capabilities for motion of weak targets in satellite videos, and insufficient training data leading to poor generalization. The Timesformer approach, illustrated in Figs. 7(d) and 8(d), demonstrates balanced but suboptimal performance across both satellite videos. While it achieves moderate detection capabilities in both maritime and terrestrial scenarios, its overall detection precision falls below acceptable thresholds for practical applications. TANet, illustrated in Figs. 7(e) and 8(e), shows initial promise in satellite video 1, achieving satisfactory results. However, its performance markedly deteriorates in satellite video 2, revealing a lack of robustness and cross-scenario generalization capabilities.

The visualized results of the proposed method, depicted in Figs. 7(a) and 8(a), demonstrate its capacity to accurately identify EOIs across a variety of environments, encompassing both maritime and terrestrial settings. The images clearly illustrate how the method effectively isolates key regions from large satellite video scenes, focusing specifically on areas that exhibit significant activities, such as boat movements in maritime regions or vehicles traversing roads and intersections on land. By accurately distinguishing these events of interest, the proposed approach significantly reduces the computational burden of processing irrelevant background areas, which otherwise hold no practical value for specific applications. This selective filtering enables a more efficient focus on the relevant parts of satellite video scenes, which is crucial for subsequent tasks like target detection, tracking, and monitoring. The red and yellow arrows vividly demonstrate the model's ability to capture the motion trajectories of boats and vehicles, offering predictive insights into target movement trends. For instance, the detected vehicle movement patterns can be utilized to predict traffic flow dynamics and identify potential congestion zones, while maritime vessel trajectories can inform maritime traffic management systems.

V. Conclusion

In this article, we have proposed a novel framework for detecting EOIs in satellite video data, addressing the unique challenges posed by its large spatial scale, sparse motion information, and significant foreground–background imbalance. By leveraging spatiotemporal slice representation, the proposed method transforms complex 3-D motion information into 2-D motion trajectories, preserving essential dynamic features while eliminating background redundancy. The integration of a ResNet-12 network for feature extraction and a custom EMD metric has demonstrated enhanced detection precision and robustness. Extensive experimental results validate the framework's superior performance compared to traditional methods, highlighting its ability to efficiently and accurately distinguish EOIs from non-EOIs in satellite video datasets. The proposed approach not only addresses the computational and analytical challenges inherent to satellite video processing but also demonstrates significant potential for real-world applications, including traffic monitoring, disaster management, and environmental observation.

Future research should focus on two critical directions to advance this framework. First, integrating the proposed method with advanced satellite platforms could enable on-board processing capabilities, allowing near real-time analytics directly at the data source. This integration would significantly reduce latency and enhance decision-making efficiency, particularly in time-sensitive applications such as security monitoring. Second, incorporating semi-supervised or unsupervised learning techniques could address the challenge of limited labeled data, improving the model's adaptability and generalizability across diverse datasets and application scenarios.

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